

1 Graph Signal Processing

Although details vary from case to case, a typical EEG processing pipeline can be described by the following sequence of steps [30]: signal acquisition, band-pass filtering, downsampling, artifact rejection, and quantification. The quantification step depends on the task at hand and may include segmentation, averaging, frequency analysis, and statistical analysis. As we will see, the incorporation of GSP into this pipeline will benefit several of these steps.

The foundations of GSP can be traced back to Püschel and Moura's work on algebraic signal processing in 2003 and the work of several researchers on graph Laplacian spectral clustering and image processing. For a detailed historical perspective, see section B of Ortega et al. [26]. In this work, we use the setting defined by Shuman [31], which has the advantage that generalizes the standard finite length discrete time signal processing case.¹ This setting considers a weighted² graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, W\}$, where $\mathcal{V}$ is the set of vertices (with cardinality $|\mathcal{V}| = N$), $\mathcal{E}$ is a set of edges, and W is a weighted adjacency matrix. The cornerstone of GSP is the graph Laplacian matrix $\mathcal{L} = D - W$,³ where D is the diagonal degree matrix with entry d_i equals to the sum of the weights incident to vertex i [32]. Since the graph Laplacian matrix is symmetric with real entries, it has non-negative eigenvalues. We denote this set of eigenvalues as $\sigma(\mathcal{L}) = \{\lambda_0, \dots, \lambda_{N-1}\}$.

The set $\sigma(\mathcal{L})$ is called the spectrum of the graph Laplacian. One of the properties of the graph Laplacian is that its smallest eigenvalue is 0 [32]. We also assume that the eigenvalues are ordered. Thus, we can write $0 = \lambda_0 \leq \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_{N-1}$.

Also, because the graph Laplacian matrix is symmetric with real entries, its eigenvectors are orthonormal [32]. Although the eigenvectors are not unique, we fix them for a given problem. We denote the set of eigenvectors by $\{u_\ell\}_{\ell=0,1,\dots,N-1}$.

The eigenvectors of the graph Laplacian provide the notion of frequency, with eigenvectors associated with larger eigenvalues corresponding to higher frequencies [33]. For instance, for any graph, the first eigenvector u_0 with eigenvalue $\lambda_0 = 0$ is always a constant. Thus, this eigenvector matches the concept of a continuous or DC component. Another example useful to illustrate the relationship between the eigenvectors of the graph Laplacian and the concept of frequency is the case of the path (or line) graph. In this graph, vertex i is connected only to vertex $i - 1$ and $i + 1$ (except vertex 0 and $N - 1$, which are connected only to vertex 1 and $N - 2$, respectively). The eigenvectors of this graph are sinusoidal, with frequencies increasing with the corresponding eigenvalue [34–36]. A signal defined over a path graph matches the standard finite-length signal considered in classical signal processing. Thus, the sinusoidal nature of the eigenvectors of the path graph Laplacian allows one to consider GSP as a generalization of the classic signal processing framework.

These properties of the graph Laplacian allow the definition of a *graph Fourier transform* for a signal $f : \mathcal{V} \rightarrow \mathbb{R}$ defined over the vertices of the graph. The graph Fourier transform of this signal is $\hat{f}(\lambda_\ell) = \sum_{i=0}^{N-1} f(i)u_\ell^*(i)$. And the corresponding inverse graph Fourier is given by $f(i) = \sum_{\ell=0}^{N-1} \hat{f}(\ell)u_\ell(i)$.

The justification for this definition is based on the following fact: The classical Fourier transform expands the signal in terms of complex exponentials, which are eigenfunctions of the one-dimensional

¹Under this framework, one gets a finite length discrete time signal with a signal defined over an unweighted line graph.

²The case of an unweighted graph can be handle as a particular case with all the weights set to one.

³It is also possible to use the normalized graph Laplacian $\tilde{\mathcal{L}} = D^{-1/2}\mathcal{L}D^{-1/2}$. See Shuman et al. [31] for a discussion about the implications of using this matrix instead of the graph Laplacian.

Laplace operator. Similarly, the graph Fourier transform expands the signal in terms of eigenvalues of the graph Laplacian matrix, a similar operator (hence the name of this matrix) [31].

Another critical ingredient of GSP is the concept of *shift* or *delay*. In classical signal processing, sample shifts are used to characterize both signals and systems (filters). For this reason, it is crucial to carry this concept to the GSP framework. By generalizing the mechanism of a shift operation of a line graph (for which GSP coincides with classical signal processing), one can define the *graph shift* operation as [26, 37] $\tilde{f} = Wf$. This definition of shifting corresponds to replacing the value of the function at a given vertex by the weighted sum of its neighbor values. A matrix-vector multiplication can describe the operation of a linear filter over a signal f on a graph: $f_{out} = Hf$, where H is an $N \times N$ *filter matrix*, and f_{out} is the output of the filter. A graph filter must be shift-invariant to have properties similar to standard filters: $W(Hf) = H(Wf)$. In words, this means that it is the same to filter the signal and then shift it than to shift the signal and then filter it. The filter matrix H satisfies this shift-invariant property if and only if there exists a polynomial $h(x) = h_0 + h_1x + \dots + h_Lx^L$ such that $H = h_0I + h_1W + \dots + h_LW^L$, where L is the order of the filter, and I is the identity matrix of size L [37]. The coefficients h_i of polynomial $h(x)$ are analogous to the filter taps of a classic Finite Impulse Response (FIR) filter.

Alternatively, graph signal f can be filtered in the frequency domain by multiplying its spectrum $\hat{f}(\lambda_\ell)$ by the transfer function $\hat{h}(\lambda_\ell)$ and then taking the inverse graph Fourier transform [31]: $f_{out} = \sum_{i=0}^{N-1} \hat{f}(\lambda_\ell) \hat{h}(\lambda_\ell) u_\ell(i)$. For more details about the properties and design criteria of graph filters, see [26, 31, 37] and references therein.

Base on these core operations—Fourier decomposition, shifting, and filtering—many of the classic signal processing operations have been extended to the GSP framework. These operations include wavelet analysis [38–43], regularization [44–48], sampling [49–54], and interpolation [55–57]. Researchers have used these techniques in many applications: In sensor networks, GSP can detect outliers, filter the data, and reconstruct sensor readings from sparse measurements [58–63]. In applications where the graph is intrinsic to the data source, such as in urban air pollution, electrical power consumption, and transportation networks, GSP can be used to find new patterns and events [64–67]. Ortega et al. [26] present a thorough revision of applications of GSP, including applications in image processing and machine learning. The following subsection describes how GSP has been used in neuroimaging problems in general and EEG in particular. It also describes how this proposal is novel with respect to the state of the art.

2 GSP and EEG

There is a long tradition of using graph theory in neuroimaging in general [28] and EEG in particular [68, 69]. However, it is essential to highlight that this use of graph theory has been circumscribed to analyze the graph obtained from connectivity structures in the brain after applying other processing steps. In this respect, GSP—that considers the analysis of *signals* defined over a graph—is fundamentally different from this classic approach. Ultimately, this extended point of view will generate more precise results and newer insights.

The capability of GSP to deal with signals over graphs has been exploited in several works related to neuroimaging. In general, these works are based on representing the interconnection among brain regions as a graph. In this graph, each brain region is a vertex. The connectivity between vertices is inferred through the anatomical connectivity or the functional relationship of the regions [70, 71]. The signal over the graph is the activity level of each region, determined typically by

functional Magnetic Resonance Imaging (fMRI) or similar techniques. For example, in [72–74], the notion of graph signal frequency is used to characterize adaptability levels and exposure to a task in a subject, using fMRI. Another example is using graph wavelet analysis to find patterns of brain activity and to decompose fMRI data [75–77]. Using GSP, it is also possible to combine fMRI with diffusion MRI in a multilayer-graph framework that captures transient networks of spatio-temporal connectivity [78]. As a final example, we mention the use of GSP to classify brain graph signals to understand the etiology of mental diseases [79–82].

Presently, most uses of GSP are based mainly on fMRI data [72,83] and the corresponding graph signal induced by the functional and anatomical connectivity. There are a limited number of works that use GSP with EEG data. In [84], the authors use GSP to detect epilepsy from the EEG. In this work, the graph is constructed from the time series data using a visibility graph. In [85], the authors also built the graph from the time series data and showed how using GSP for dimensionality reduction improves the classification performance over classic techniques. Other works used GSP for noise suppression, dimensionality reduction, and classification [86–88]. All these works ignore geometrical information to build the graph. To the best of our knowledge, the work of Mortaheb et al. [89] is the only one that uses geometric information by considering the electrodes’ distance to build the graph. GSP is then used to compute the total variation of the graph signal to characterize disorders of consciousness.

Based on the literature review we presented in the previous paragraphs, we can conclude that **this proposal will be the first work to use GSP extensively to process EEG at the sensor (electrode) level.** Working at the sensor level is essential because, in a clinical setting, fMRI or EEG source activity data are rarely available, and the clinician has to make a diagnostic decision based on inspection of the scalp data alone. We expect this novel approach to bringing new perspectives and tools to deal with the challenges that have emerged during the last years. The proposal will contribute to all the steps of the EEG processing pipeline.

To illustrate how GSP can be used in the EEG processing pipeline, we now show a preliminary experiment where we use GSP to impute missing sensor data. Data imputation is a common situation in practice that happens when the skin-electrode connection is inappropriate. We use the sample dataset provided by the MNE-Python library [90]. This dataset consists of a 60-channel electrode recording of the EEG (with one electrode as reference), while the subject was presented with different visual and auditory stimuli.⁴ We computed the Event-Related Potential (ERP) for the left auditory condition at 100 milliseconds using the average reference (Figure 1-(a)). We consider each electrode as a graph vertex and the value of the ERP at 100 milliseconds on each electrode as the graph signal. For each vertex, we add an edge between this vertex and its k -nearest neighbors (based on the Euclidean distance of the position of the electrodes). Figures 1-(b) and 1-(c) show the corresponding adjacency matrix of this graph and a plot of the graph and the graph signal, respectively, for a graph built using the $k = 5$ nearest neighbors.

In this experiment, we assume that data from electrode 6 (marked with a red circle in figure 1-(c)) is missing. Then, we use GSP to impute the missing value based on the remaining electrodes’ signal value. In particular, we do data imputation by solving the graph linear regression problem [31,91]

$$\begin{aligned} \hat{f} &= \arg \min_x x^T \mathcal{L} x \\ \text{s.t. } f_M &= Mx, \end{aligned}$$

where $\hat{f}$ is the graph signal we wish to estimate, $\mathcal{L}$ is the graph Laplacian, f is the graph signal

⁴See more details about the dataset in <https://bit.ly/3dzZnHo>.

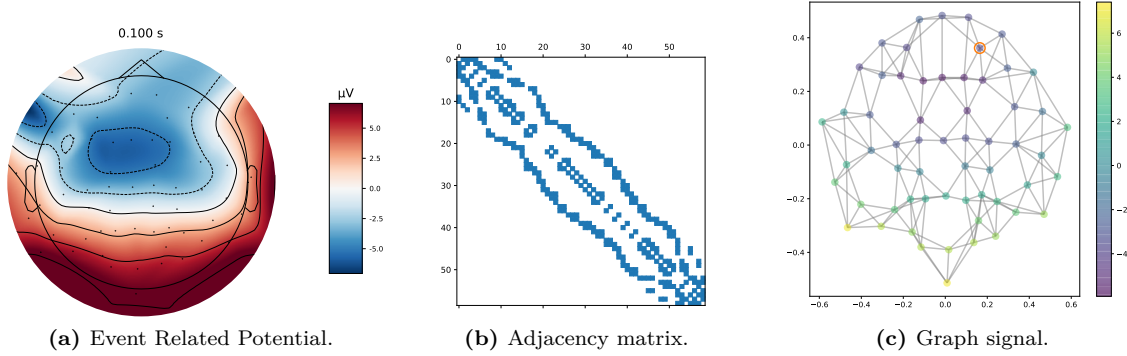


Figure 1: A graph is built using each electrode as a vertex and an edge between each vertex and its five nearest neighbors. The ERP at each electrode (vertex of the graph) is the graph signal: (a) Topography of the ERP at 100 milliseconds. (b) The adjacency matrix of the graph. (c) Graph and signal graph. The red circle marks the electrode with missing data.

with missing data, f_M is a vector with the non-missing values, and M is a masking matrix that extracts the non-missing values.

To evaluate the performance of the GSP-based imputation method, we computed the magnitude of the error for different values of numbers of neighbors used to build the graph. We compared this error with the error obtained by imputing the missing data using a standard Radial Basis Function (RBF) interpolator [92]. Figure 2-(a) shows the result. This result highlights two facts: (a) For a reasonably build graph, GSP-based data imputation is better than the standard RBF imputation. By reasonably, we mean a graph built with a relatively small number of neighbors. This characteristic makes sense, since at some point, by increasing the number of neighbors k , the GSP imputation begins to consider vertices that are too far away (in a Euclidean distance sense). (b) This effect is amplified by the fact that we are using an unweighted graph in this example, highlighting the relevance of the graph topology on the performance of GSP methods.

To study the effect of using a weighted graph we repeat the experiment using a graph built using the thresholded Gaussian kernel (see Eq. (1)). We computed the magnitude of the error for different values of the dispersion parameter θ of the kernel. Figure 2-(b) shows the result. As a reference, we show in the figure the RBF error (red dashed line) and the smallest error obtained with the unweighted graph (purple dashed line). We observe that by using a weighted graph it is possible to obtain errors smaller than with the unweighted graph. We also observe that the dependency of the error with respect to the kernel scaling parameter is more robust. These results highlight the importance of the graph used for the analysis. For this reason, the best methods to build graphs from a given set of EEG sensors will be investigated in detail during the execution of the proposal (see following sections).

3 Goals, hypotheses, and research questions

The main goal of this project is to leverage Graph Signal Processing to improve the processing and understanding of the electroencephalogram.

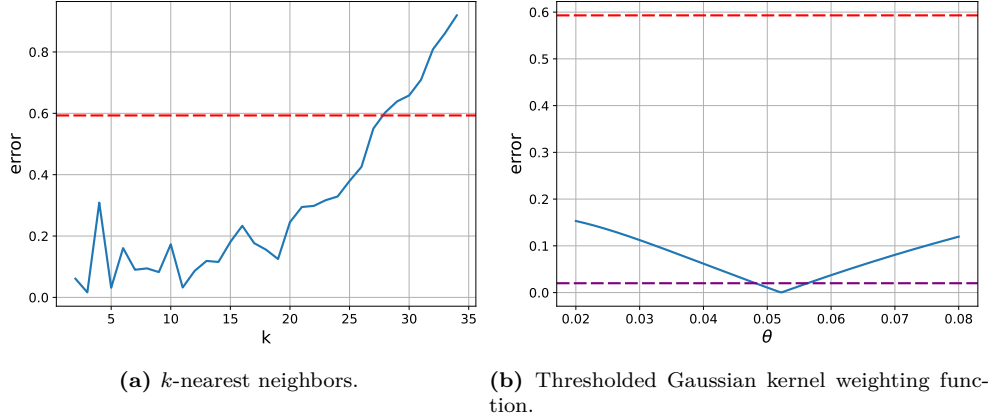


Figure 2: Magnitude of the imputation error. (a) With an unweighted graph built using the k -nearest neighbors as a function of k . (b) With a weighted graph built using a thresholded Gaussian kernel as a function of θ .

We organize the proposal along the following specific aims:

Specific Aim 1 (SA1): *To improve classic EEG pre-processing using GSP.* Most classic pre-processing EEG techniques ignore the intrinsic graph structure of the data. For this reason, we expect that, by introducing GSP into these techniques, we will observe improvements in the performance of these methods. Examples of the methods that we will study include filtering, downsampling, and artifact rejection (such as GSP use to improve Independent Component Analysis (ICA) artifact rejection). This specific aim also includes the study of how to build the graph effectively.

A corresponding hypothesis supports each specific aim of this proposal:

Hypothesis 1 (H1): *EEG filtering, downsampling, and artifact rejection methods based on GSP have better performance than their classic counterparts.*

4 Methodology

SA1: To improve classic EEG pre-processing using GSP. The first step for using GSP is to construct a proper graph. As we showed in the previous experiment, GSP’s performance depends on how we define the edge set of the graph. Here, following the ideas of our preliminary experiment, we will build a weighted graph using a thresholded Gaussian kernel [31] where the weights between vertices i and j are given by

$$W_{i,j} = \begin{cases} \exp\left(-\frac{\text{dist}^2(i,j)}{2\theta^2}\right) & \text{if } \text{dist}(i,j) \leq \kappa \\ 0 & \text{otherwise,} \end{cases} \quad (1)$$

where κ is the threshold, θ is a dispersion factor, and $\text{dist}(i, j)$ is the distance—euclidean or geodesical—between vertices i and j . We will evaluate the effect of these parameters by comparing their impact on the performance of different processing tasks, such as missing data imputation, in a similar vein to the example of section 1. As an alternative, we will build the graph using a data-based approach based on the estimation of covariance matrices [100, 101]. Significant differences between these two types of graphs will, for instance, allow us to identify bad electrodes automatically. We will quantify the similarity between these two graphs using the *graph edit distance* (GED) [102] and the *centrality distance* [103]. We will then find an appropriate threshold for these similarity measures following an approach based on cross-validation and Bayes optimization (see [15] for more details).

It is essential to notice that we are assuming that the graph does not change over time. We justify this assumption by stating that the graph structure depends on the spatial relationship between the electrodes. Thus, even if one considers brain dynamics, that dynamic will affect the signal, but not the graph. Of course, this argument does not hold for the graph built using the covariance matrix estimation. However, we use this last technique to detect inconsistencies in the data and not as the primary strategy to build the graph. In any case, this is an aspect that we will consider when using the data-driven approach to build the graph.

After knowing the most effective way to build graphs for the sensor network formed by the EEG electrodes, we will extend classic processing algorithms, such as filtering and ICA, using GSP. These new techniques will be evaluated with criteria similar to those previously used in the literature [15].

Evaluating the performance of the proposed methods with real data is challenging since there is no clear definition of ground truth. To evaluate the performance of the method for identifying bad electrodes, we will use two public datasets where bad electrodes were manually labeled: The MNE sample data set [90] (with one subject) and the EEGBCI dataset [104] (with 109 subjects). We will consider the manually cleaned data as ground truth, and we will evaluate the performance of the method using [15] $\|X(\text{method}) - X(\text{clean})\|_\infty$, where $X(\text{method})$ is a matrix with the evoked potentials computed after removing the bad electrodes with the method, and $X(\text{method})$ is a matrix with the evoked potentials computed using the data manually cleaned. We will compare its performance against other bad channel detection methods, such as Autoreject, FASTER, SNS, and RANSAC [15]. Since we are interested in applying this method in large datasets, we will also evaluate our approach’s execution time and compare it against the execution time of these same methods.

To evaluate GSP-based artifact rejection method performance, we will mix a known artifact with an artifact-free signal, and then we will compute the correlation between the original artifact and the one extracted by the method [105, 106]. We will compare this correlation with the one obtained using standard methods such as ICA.

The methods developed in the framework of this specific aim will contribute to the set of techniques available to deal with large EEG datasets and solve reproducibility issues that need to avoid manual interventions [15, 107].

We will also use GSP to develop new electrode placement techniques. To this end, we will use the smooth graph signal model one obtains by appropriately building the graph to build effective reconstruction methods from a small number of sensors [26]. This signal model can be combined with recent developments in sampling on graphs theory [108] to develop new optimal sensor placement strategies. We believe this new technique has a relevant clinical application since, in these settings, it is vital to have a very low number of electrodes. For this problem, we start with a set of standard high-density electrode locations—such as the international 10-20 system—and then select a subset

of electrode locations, denoted $\mathcal{T}$, by finding the corresponding subset of the graph with a graph Laplacian with the largest minimum eigenvalue [108, 109]. This selection process is equivalent to solve the optimization problem

$$\max_{\mathcal{T}} \lambda_{\min}(\mathcal{L}_{\mathcal{T}}^T \mathcal{L}_{\mathcal{T}}),$$

where $\mathcal{L}_{\mathcal{T}}$ is the graph Laplacian matrix corresponding to the graph generated by the electrodes selected by the set $\mathcal{T}$. We will extend this sensor placement optimization method to incorporate constraints on the subset $\mathcal{T}$ to focus on specific features relevant to the given task, as the **Mu Rhythm for a BCI task**.

To evaluate the performance of the developed sensor placement techniques, we will consider the high-density EEG dataset as ground truth. We will then run a typical EEG analysis using all the electrodes and the selected electrodes, and then we will compare the results. We will consider different analyses typically performed in BCI and clinical problems since these are applications requiring the use of a small number of electrodes. For instance, we will solve a motor imagery decoding problem—where one uses a classifier to predict what the subject imagined based on the recorded EEG—using the “EEG Motor Movement/Imagery Dataset” [104] with features extracted using the Common Spatial Pattern (CSP) [110] and a Linear Discriminant Analysis (LDA) classifier.⁵ We will compare the classification accuracy using: all 64 electrodes available in the dataset, a small subset of electrodes commonly used for BCI, and a subset of electrodes selected using the GSP method.

Datasets: Since one of the main focuses of the proposal is on developing new processing algorithms and techniques, we will perform all the experiments using public EEG datasets. As detailed above, we are using five datasets: (1) The MNE sample data set [90]; (2) the EEGBCI dataset [104]; (3) The EEG Motor Movement/Imagery Dataset [104]; (4) The IVa BCI Competition III dataset [115]; and (5) the TDBRAIN database [116]. We chose these datasets because each one has features—such as number of subjects, number of electrodes, experimental conditions, etc.—that will allow us to evaluate the specific characteristics of the developed techniques. We claim that this approach is more efficient and effective since it allows us to focus our efforts on reaching the proposal’s goals. In addition to the datasets already mentioned, some examples of datasets that we will also may use include “A dataset for the test-retest reliability assessment of EEG & ERP quantities” [121] and the “EEG time series from the Department of Epileptology at the University Hospital of Bonn” [122].⁶

Code availability: All the numerical methods and algorithms will be packed as a free software library. This library will be available to the public under a liberal license, such as BSD or similar. The software will be compatible with the API of MNE [90] and PyGSP [123]. In this proposal, we consider one programmer’s participation as a member of the technical staff, dedicated to packaging all the algorithms in this library, including aspects such as documentation, unit testing, and continuous integration.

⁵See https://mne.tools/stable/auto_examples/decoding/plot_decoding_csp_eeg.html for an example of how to perform this decoding task.

⁶See <https://github.com/meagmohit/EEG-Datasets> for a comprehensive list of EEG datasets that we will consider in this proposal.

5 Work Plan

The duration of the proposal is three years, with one year allocated with each specific aim. The following are the tasks and milestones associated with each specific aim:

SA1: To improve classic EEG pre-processing using GSP.

Task 1: Development of graph construction techniques. This task includes the analysis of the construction techniques for the graph for using the thresholded Gaussian kernel approach and the data-driven approach.

Task 2: Development of electrode placement techniques. In this task we will use smooth graph signal models to build reconstruction methods based on a small number of electrodes.

Task 3: Extension of classic EEG processing algorithms using GSP. Based on the graph construction techniques developed in the previous task, we will work in the extensions and validation of classic signal processing techniques using the GSP domain.

Task 4: Packaging of the developed algorithms into a software library. All the techniques and developed algorithms will be transform into a Python library available to the community. The library will include the corresponding testing and documentation.

Task 5: Manuscript preparation and submission. We will send the manuscript to Neuroimage, Frontiers in Human Neuroscience, Brain Topography, or similar.

References

- [1] A. F. Jackson and D. J. Bolger, “The neurophysiological bases of EEG and EEG measurement: a review for the rest of us,” *Psychophysiology*, vol. 51, no. 11, pp. 1061–1071, Nov. 2014.
- [2] A. J. Casson, M. Abdulaal, M. Dulabh, S. Kohli, S. Krachunov, and E. Trimble, “Electroencephalogram,” in *Seamless Healthcare Monitoring*. Springer, Cham, 2018, pp. 45–81.
- [3] E. Niedermeyer and F. L. da Silva, *Electroencephalography: basic principles, clinical applications, and related fields*. Lippincott Williams & Wilkins, 2005.
- [4] S. Tong and N. V. Thakor, *Quantitative EEG analysis methods and clinical applications*. Artech House, 2009.
- [5] C. Rubiños and D. A. Godoy, “Electroencephalographic monitoring in the critically ill patient: What useful information can it contribute?” *Medicina Intensiva (English Edition)*, vol. 44, no. 5, pp. 301–309, Jun. 2020.
- [6] J. T. Cacioppo, L. G. Tassinary, and G. Berntson, *Handbook of psychophysiology*. Cambridge university press, 2007.
- [7] H. Abbood, W. Al-Nuaimy, A. Al-Ataby, S. A. Salem, and H. S. AlZubi, “Prediction of driver fatigue: Approaches and open challenges,” in *2014 14th UK Workshop on Computational Intelligence (UKCI)*, Sep. 2014, pp. 1–6, iSSN: 2162-7657.
- [8] G. A. M. Vasiljevic and L. C. d. Miranda, “Brain–Computer Interface Games Based on Consumer-Grade EEG Devices: A Systematic Literature Review,” *International Journal of Human–Computer Interaction*, vol. 36, no. 2, pp. 105–142, Jan. 2020, publisher: Taylor & Francis _eprint: <https://doi.org/10.1080/10447318.2019.1612213>.
- [9] S. J. Luck, *An introduction to the event-related potential technique*. MIT press, 2014.
- [10] M. X. Cohen, *Analyzing neural time series data: theory and practice*. MIT press, 2014.
- [11] R. Hari and A. Puce, *MEG-EEG Primer*. Oxford University Press, 2017.

- [12] M. Weiergräber, A. Papazoglou, K. Broich, and R. Müller, “Sampling rate, signal bandwidth and related pitfalls in EEG analysis,” *Journal of Neuroscience Methods*, vol. 268, pp. 53–55, Aug. 2016.
- [13] M. J. Larson and K. A. Carbine, “Sample size calculations in human electrophysiology (EEG and ERP) studies: A systematic review and recommendations for increased rigor,” *International Journal of Psychophysiology*, vol. 111, pp. 33–41, Jan. 2017.
- [14] D. Yao, Y. Qin, S. Hu, L. Dong, M. L. Bringas Vega, and P. A. Valdés Sosa, “Which Reference Should We Use for EEG and ERP practice?” *Brain Topogr*, vol. 32, no. 4, pp. 530–549, Jul. 2019.
- [15] M. Jas, D. A. Engemann, Y. Bekhti, F. Raimondo, and A. Gramfort, “Autoreject: Automated artifact rejection for MEG and EEG data,” *NeuroImage*, vol. 159, pp. 417–429, Oct. 2017.
- [16] A. Barachant, A. Andreev, and M. Congedo, “The Riemannian Potato: an automatic and adaptive artifact detection method for online experiments using Riemannian geometry,” in *TOBI Workshop IV*, 2013, pp. 19–20.
- [17] T. P. Mutanen, J. Metsomaa, S. Liljander, and R. J. Ilmoniemi, “Automatic and robust noise suppression in EEG and MEG: The SOUND algorithm,” *NeuroImage*, vol. 166, pp. 135–151, Feb. 2018.
- [18] S. Debener and M. G. Bleichner, “Transparent electroencephalography? : Exploring ear-EEG for long-term, mobile electrophysiology,” in *2019 7th International Winter Conference on Brain-Computer Interface (BCI)*, Feb. 2019, pp. 1–2, iSSN: 2572-7672.
- [19] M. G. Bleichner and S. Debener, “Independent component decomposition of around ear EEG data to detect artifacts.” in *2019 IEEE International Conference on Systems, Man and Cybernetics (SMC)*, Oct. 2019, pp. 3631–3634, iSSN: 2577-1655.
- [20] J. Kim, J. Lee, C. Han, and K. Park, “An Instant Donning Multi-Channel EEG Headset (with Comb-Shaped Dry Electrodes) and BCI Applications,” *Sensors*, vol. 19, no. 7, p. 1537, Jan. 2019, number: 7 Publisher: Multidisciplinary Digital Publishing Institute.
- [21] S. Narayana, R. V. Prasad, and K. Warmerdam, “Mind your thoughts: BCI using single EEG electrode,” *IET Cyber-Physical Systems: Theory Applications*, vol. 4, no. 2, pp. 164–172, 2019, conference Name: IET Cyber-Physical Systems: Theory Applications.
- [22] N. Padfield, J. Zabalza, H. Zhao, V. Masero, and J. Ren, “EEG-Based Brain-Computer Interfaces Using Motor-Imagery: Techniques and Challenges,” *Sensors*, vol. 19, no. 6, p. 1423, Jan. 2019, number: 6 Publisher: Multidisciplinary Digital Publishing Institute.
- [23] V. Noreika, S. Georgieva, S. Wass, and V. Leong, “14 challenges and their solutions for conducting social neuroscience and longitudinal EEG research with infants,” *Infant Behavior and Development*, vol. 58, p. 101393, Feb. 2020.
- [24] A. Lau-Zhu, M. P. H. Lau, and G. McLoughlin, “Mobile EEG in research on neurodevelopmental disorders: Opportunities and challenges,” *Developmental Cognitive Neuroscience*, vol. 36, p. 100635, Apr. 2019.
- [25] M. A. Serhani, M. E. Menshawy, A. Benharref, S. Harous, and A. N. Navaz, “New algorithms for processing time-series big EEG data within mobile health monitoring systems,” *Computer Methods and Programs in Biomedicine*, vol. 149, pp. 79–94, Oct. 2017.
- [26] A. Ortega, P. Frossard, J. Kovačević, J. M. F. Moura, and P. Vandergheynst, “Graph Signal Processing: Overview, Challenges, and Applications,” *Proceedings of the IEEE*, vol. 106, no. 5, pp. 808–828, May 2018, conference Name: Proceedings of the IEEE.

- [27] Y.-Y. Liu, J.-J. Slotine, and A.-L. Barabási, “Controllability of complex networks,” *Nature*, vol. 473, no. 7346, pp. 167–173, May 2011, number: 7346 Publisher: Nature Publishing Group.
- [28] D. S. Bassett and O. Sporns, “Network neuroscience,” *Nat Neurosci*, vol. 20, no. 3, pp. 353–364, Mar. 2017.
- [29] S. M. Smith, D. Vidaurre, C. F. Beckmann, M. F. Glasser, M. Jenkinson, K. L. Miller, T. E. Nichols, E. C. Robinson, G. Salimi-Khorshidi, M. W. Woolrich, D. M. Barch, K. Uğurbil, and D. C. Van Essen, “Functional connectomics from resting-state fMRI,” *Trends in Cognitive Sciences*, vol. 17, no. 12, pp. 666–682, Dec. 2013.
- [30] S. Sanei and J. A. Chambers, *EEG signal processing*. John Wiley & Sons, 2013.
- [31] D. I. Shuman, S. K. Narang, P. Frossard, A. Ortega, and P. Vandergheynst, “The emerging field of signal processing on graphs: Extending high-dimensional data analysis to networks and other irregular domains,” *IEEE Signal Processing Magazine*, vol. 30, no. 3, pp. 83–98, May 2013, conference Name: IEEE Signal Processing Magazine.
- [32] A. E. Brouwer and W. H. Haemers, *Spectra of graphs*. Springer Science & Business Media, 2011.
- [33] A. Sandryhaila and J. M. F. Moura, “Discrete Signal Processing on Graphs: Frequency Analysis,” *IEEE Transactions on Signal Processing*, vol. 62, no. 12, pp. 3042–3054, Jun. 2014, conference Name: IEEE Transactions on Signal Processing.
- [34] D. Spielman, “Spectral graph theory,” in *Combinatorial scientific computing*. Citeseer, 2012, no. 18.
- [35] D. A. Spielman, “Spectral graph theory,” 2009.
- [36] F. R. Chung and F. C. Graham, *Spectral graph theory*. American Mathematical Soc., 1997, no. 92.
- [37] A. Sandryhaila and J. M. Moura, “Discrete signal processing on graphs,” *IEEE transactions on signal processing*, vol. 61, no. 7, pp. 1644–1656, 2013, publisher: IEEE.
- [38] R. R. Coifman and M. Maggioni, “Diffusion wavelets,” *Applied and Computational Harmonic Analysis*, vol. 21, no. 1, pp. 53–94, 2006.
- [39] D. K. Hammond, P. Vandergheynst, and R. Gribonval, “Wavelets on graphs via spectral graph theory,” *Applied and Computational Harmonic Analysis*, vol. 2, Apr. 2010.
- [40] W. Wang and K. Ramchandran, “Random multiresolution representations for arbitrary sensor network graphs,” in *2006 IEEE International Conference on Acoustics Speech and Signal Processing Proceedings*, vol. 4. IEEE, 2006, pp. IV–IV.
- [41] A. D. Szlam, M. Maggioni, R. R. Coifman, and J. C. Bremer Jr, “Diffusion-driven multiscale analysis on manifolds and graphs: top-down and bottom-up constructions,” in *Wavelets XI*, vol. 5914. International Society for Optics and Photonics, 2005, p. 59141D.
- [42] J. C. Bremer, R. R. Coifman, M. Maggioni, and A. D. Szlam, “Diffusion Wavelet Packets,” *Applied and Computational Harmonic Analysis*, vol. 21, no. 2006, pp. 95–112, 2006.
- [43] M. Maggioni, J. C. Bremer Jr, R. R. Coifman, and A. D. Szlam, “Biorthogonal diffusion wavelets for multiscale representation on manifolds and graphs,” in *Wavelets XI*, vol. 5914. International Society for Optics and Photonics, 2005, p. 59141M.
- [44] D. Zhou and B. Schölkopf, “Regularization on discrete spaces,” in *Joint Pattern Recognition Symposium*. Springer, 2005, pp. 361–368.
- [45] G. Peyré, S. Bogleux, and L. Cohen, “Non-local regularization of inverse problems,” in *European Conference on Computer Vision*. Springer, 2008, pp. 57–68.

- [46] F. Zhang and E. R. Hancock, "Graph spectral image smoothing using the heat kernel," *Pattern Recognition*, vol. 41, no. 11, pp. 3328–3342, 2008, publisher: Elsevier.
- [47] S. Bogleux, A. Elmoataz, and M. Melkemi, "Discrete regularization on weighted graphs for image and mesh filtering," in *International Conference on Scale Space and Variational Methods in Computer Vision*. Springer, 2007, pp. 128–139.
- [48] A. Elmoataz, O. Lezoray, and S. Bogleux, "Nonlocal discrete regularization on weighted graphs: a framework for image and manifold processing," *IEEE transactions on Image Processing*, vol. 17, no. 7, pp. 1047–1060, 2008, publisher: IEEE.
- [49] I. Pesenson, "Sampling in Paley-Wiener spaces on combinatorial graphs," *Transactions of the American Mathematical Society*, vol. 360, no. 10, pp. 5603–5627, 2008.
- [50] A. Anis, A. Gadde, and A. Ortega, "Towards a sampling theorem for signals on arbitrary graphs," in *2014 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2014, pp. 3864–3868.
- [51] H. Shomorony and A. S. Avestimehr, "Sampling large data on graphs," in *2014 IEEE Global Conference on Signal and Information Processing (GlobalSIP)*. IEEE, 2014, pp. 933–936.
- [52] S. Chen, R. Varma, A. Sandryhaila, and J. Kovačević, "Discrete Signal Processing on Graphs: Sampling Theory," *IEEE transactions on signal processing*, vol. 63, no. 24, pp. 6510–6523, 2015, publisher: IEEE.
- [53] A. Anis, A. Gadde, and A. Ortega, "Efficient sampling set selection for bandlimited graph signals using graph spectral proxies," *IEEE Transactions on Signal Processing*, vol. 64, no. 14, pp. 3775–3789, 2016, publisher: IEEE.
- [54] S. Chen, A. Sandryhaila, J. M. Moura, and J. Kovačević, "Signal recovery on graphs: Variation minimization," *IEEE Transactions on Signal Processing*, vol. 63, no. 17, pp. 4609–4624, 2015, publisher: IEEE.
- [55] S. K. Narang, A. Gadde, E. Sanou, and A. Ortega, "Localized iterative methods for interpolation in graph structured data," in *2013 IEEE Global Conference on Signal and Information Processing*. IEEE, 2013, pp. 491–494.
- [56] S. K. Narang, A. Gadde, and A. Ortega, "Signal processing techniques for interpolation in graph structured data," in *2013 IEEE International Conference on Acoustics, Speech and Signal Processing*. IEEE, 2013, pp. 5445–5449.
- [57] X. Wang, P. Liu, and Y. Gu, "Local-set-based graph signal reconstruction," *IEEE Transactions on Signal Processing*, vol. 63, no. 9, pp. 2432–2444, 2015, publisher: IEEE.
- [58] R. Wagner, H. Choi, R. Baraniuk, and V. Delouille, "Distributed wavelet transform for irregular sensor network grids," in *IEEE/SP 13th Workshop on Statistical Signal Processing, 2005*. IEEE, 2005, pp. 1196–1201.
- [59] R. S. Wagner, R. G. Baraniuk, S. Du, D. B. Johnson, and A. Cohen, "An architecture for distributed wavelet analysis and processing in sensor networks," in *Proceedings of the 5th international conference on Information processing in sensor networks*, ser. IPSN '06. Nashville, Tennessee, USA: Association for Computing Machinery, Apr. 2006, pp. 243–250.
- [60] A. Ciancio, S. Patten, A. Ortega, and B. Krishnamachari, "Energy-efficient data representation and routing for wireless sensor networks based on a distributed wavelet compression algorithm," in *Proceedings of the 5th international conference on Information processing in sensor networks*, ser. IPSN '06. Nashville, Tennessee, USA: Association for Computing Machinery, Apr. 2006, pp. 309–316.

- [61] G. Shen and A. Ortega, "Joint Routing and 2D Transform Optimization for Irregular Sensor Network Grids Using Wavelet Lifting," in *2008 International Conference on Information Processing in Sensor Networks (ipsn 2008)*, Apr. 2008, pp. 183–194.
- [62] X. Zhu and M. Rabbat, "Graph Spectral Compressed Sensing for Sensor Networks," in *submitted to the 2012 IEEE International Conference on Acoustics, Speech, and Signal Processing*, 2012.
- [63] M. Kaneko, G. Cheung, W.-T. Su, and C.-W. Lin, "Graph-Based Joint Signal/Power Restoration for Energy Harvesting Wireless Sensor Networks," in *GLOBECOM 2017 - 2017 IEEE Global Communications Conference*, Dec. 2017, pp. 1–6.
- [64] R. K. Jain, J. M. Moura, and C. E. Kontokosta, "Big Data + Big Cities: Graph Signals of Urban Air Pollution [Exploratory SP]," *IEEE Signal Processing Magazine*, vol. 31, no. 5, pp. 130–136, Sep. 2014, conference Name: IEEE Signal Processing Magazine.
- [65] P. Valdivia, F. Dias, F. Petronetto, C. T. Silva, and L. G. Nonato, "Wavelet-based visualization of time-varying data on graphs," in *2015 IEEE Conference on Visual Analytics Science and Technology (VAST)*, Oct. 2015, pp. 1–8.
- [66] X. Dong, A. Ortega, P. Frossard, and P. Vandergheynst, "Inference of mobility patterns via Spectral Graph Wavelets," in *2013 IEEE International Conference on Acoustics, Speech and Signal Processing*, May 2013, pp. 3118–3122, iSSN: 2379-190X.
- [67] D. M. Mohan, M. T. Asif, N. Mitrovic, J. Dauwels, and P. Jaillet, "Wavelets on graphs with application to transportation networks," in *17th International IEEE Conference on Intelligent Transportation Systems (ITSC)*, Oct. 2014, pp. 1707–1712, iSSN: 2153-0017.
- [68] M. N. Anastasiadou, M. Christodoulakis, E. S. Papathanasiou, S. S. Papacostas, A. Hadjipapas, and G. D. Mitsis, "Graph Theoretical Characteristics of EEG-Based Functional Brain Networks in Patients With Epilepsy: The Effect of Reference Choice and Volume Conduction," *Front. Neurosci.*, vol. 13, 2019, publisher: Frontiers.
- [69] M. Kaminski and K. J. Blinowska, "Is Graph Theoretical Analysis a Useful Tool for Quantification of Connectivity Obtained by Means of EEG/MEG Techniques?" *Front. Neural Circuits*, vol. 12, 2018, publisher: Frontiers.
- [70] E. Bullmore and O. Sporns, "The economy of brain network organization," *Nature Reviews Neuroscience*, vol. 13, no. 5, pp. 336–349, May 2012, number: 5 Publisher: Nature Publishing Group.
- [71] O. Sporns, *Networks of the Brain*. MIT Press, Oct. 2010.
- [72] W. Huang, L. Goldsberry, N. F. Wymbs, S. T. Grafton, D. S. Bassett, and A. Ribeiro, "Graph Frequency Analysis of Brain Signals," *IEEE Journal of Selected Topics in Signal Processing*, vol. 10, no. 7, pp. 1189–1203, Oct. 2016, conference Name: IEEE Journal of Selected Topics in Signal Processing.
- [73] L. Goldsberry, W. Huang, N. F. Wymbs, S. T. Grafton, D. S. Bassett, and A. Ribeiro, "Brain signal analytics from graph signal processing perspective," in *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Mar. 2017, pp. 851–855, iSSN: 2379-190X.
- [74] D. S. Bassett, N. F. Wymbs, M. A. Porter, P. J. Mucha, J. M. Carlson, and S. T. Grafton, "Dynamic reconfiguration of human brain networks during learning," *PNAS*, vol. 108, no. 18, pp. 7641–7646, May 2011, publisher: National Academy of Sciences Section: Biological Sciences.

- [75] H. Behjat, N. Leonardi, L. Sörnmo, and D. Van De Ville, "Anatomically-adapted graph wavelets for improved group-level fMRI activation mapping," *NeuroImage*, vol. 123, pp. 185–199, Dec. 2015.
- [76] N. Leonardi and D. Van De Ville, "Tight Wavelet Frames on Multislice Graphs," *IEEE Transactions on Signal Processing*, vol. 61, no. 13, pp. 3357–3367, Jul. 2013, conference Name: IEEE Transactions on Signal Processing.
- [77] S. Atasoy, I. Donnelly, and J. Pearson, "Human brain networks function in connectome-specific harmonic waves," *Nature Communications*, vol. 7, no. 1, p. 10340, Jan. 2016, number: 1 Publisher: Nature Publishing Group.
- [78] A. Griffa, B. Ricaud, K. Benzi, X. Bresson, A. Daducci, P. Vandergheynst, J.-P. Thiran, and P. Hagmann, "Transient networks of spatio-temporal connectivity map communication pathways in brain functional systems," *NeuroImage*, vol. 155, pp. 490–502, Jul. 2017.
- [79] M. Ménoret, N. Farrugia, B. Padeloup, and V. Gripon, "Evaluating graph signal processing for neuroimaging through classification and dimensionality reduction," in *2017 IEEE Global Conference on Signal and Information Processing (GlobalSIP)*, Nov. 2017, pp. 618–622.
- [80] C. Hu, L. Cheng, J. Sepulcre, K. A. Johnson, G. E. Fakhri, Y. M. Lu, and Q. Li, "A Spectral Graph Regression Model for Learning Brain Connectivity of Alzheimer's Disease," *PLOS ONE*, vol. 10, no. 5, p. e0128136, May 2015, publisher: Public Library of Science.
- [81] C. Hu, J. Sepulcre, K. A. Johnson, G. E. Fakhri, Y. M. Lu, and Q. Li, "Matched signal detection on graphs: Theory and application to brain imaging data classification," *NeuroImage*, vol. 125, pp. 587–600, Jan. 2016.
- [82] C. Hu, X. Hua, J. Ying, P. M. Thompson, G. E. Fakhri, and Q. Li, "Localizing Sources of Brain Disease Progression with Network Diffusion Model," *IEEE Journal of Selected Topics in Signal Processing*, vol. 10, no. 7, pp. 1214–1225, Oct. 2016, conference Name: IEEE Journal of Selected Topics in Signal Processing.
- [83] W. Huang, T. A. W. Bolton, J. D. Medaglia, D. S. Bassett, A. Ribeiro, and D. Van De Ville, "A Graph Signal Processing Perspective on Functional Brain Imaging," *Proceedings of the IEEE*, vol. 106, no. 5, pp. 868–885, May 2018, conference Name: Proceedings of the IEEE.
- [84] P. Mathur and V. K. Chakka, "Graph Signal Processing of EEG signals for Detection of Epilepsy," in *2020 7th International Conference on Signal Processing and Integrated Networks (SPIN)*, Feb. 2020, pp. 839–843, iISSN: 2688-769X.
- [85] L. Rui, H. Nejati, S. H. Safavi, and N.-M. Cheung, "Simultaneous low-rank component and graph estimation for high-dimensional graph signals: Application to brain imaging," in *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Mar. 2017, pp. 4134–4138, iISSN: 2379-190X.
- [86] L. Rui, H. Nejati, and N.-M. Cheung, "Dimensionality reduction of brain imaging data using graph signal processing," in *2016 IEEE International Conference on Image Processing (ICIP)*, Sep. 2016, pp. 1329–1333, iISSN: 2381-8549.
- [87] U. Graichen, R. Eichardt, P. Fiedler, D. Strohmeier, F. Zanow, and J. Haueisen, "SPHARA - A Generalized Spatial Fourier Analysis for Multi-Sensor Systems with Non-Uniformly Arranged Sensors: Application to EEG," *PLOS ONE*, vol. 10, no. 4, p. e0121741, Apr. 2015, publisher: Public Library of Science.
- [88] K. Smith, B. Ricaud, N. Shahid, S. Rhodes, J. M. Starr, A. Ibáñez, M. A. Parra, J. Escudero, and P. Vandergheynst, "Locating Temporal Functional Dynamics of Visual Short-Term

- Memory Binding using Graph Modular Dirichlet Energy,” *Scientific Reports*, vol. 7, no. 1, p. 42013, Feb. 2017, number: 1 Publisher: Nature Publishing Group.
- [89] S. Mortaheb, J. Annen, C. Chatelle, H. Cassol, G. Martens, A. Thibaut, O. Gosseries, and S. Laureys, “A Graph Signal Processing Approach to Study High Density EEG Signals in Patients with Disorders of Consciousness,” in *2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Jul. 2019, pp. 4549–4553, iSSN: 1558-4615.
 - [90] A. Gramfort, M. Luessi, E. Larson, D. A. Engemann, D. Strohmeier, C. Brodbeck, R. Goj, M. Jas, T. Brooks, and L. Parkkonen, “MEG and EEG data analysis with MNE-Python,” *Frontiers in neuroscience*, vol. 7, p. 267, 2013.
 - [91] G. B. Eslamlou, A. Jung, and N. Goertz, “Smooth graph signal recovery via efficient Laplacian solvers,” in *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2017, pp. 5915–5919.
 - [92] G. B. Wright, “Radial basis function interpolation: numerical and analytical developments,” phd, University of Colorado at Boulder, USA, 2003, aAI3087597.
 - [93] Y.-Y. Liu, J.-J. Slotine, and A.-L. Barabási, “Observability of complex systems,” *PNAS*, vol. 110, no. 7, pp. 2460–2465, Feb. 2013, publisher: National Academy of Sciences Section: Physical Sciences.
 - [94] S. Gu, F. Pasqualetti, M. Cieslak, Q. K. Telesford, A. B. Yu, A. E. Kahn, J. D. Medaglia, J. M. Vettel, M. B. Miller, S. T. Grafton, and D. S. Bassett, “Controllability of structural brain networks,” *Nat Commun*, vol. 6, no. 1, p. 8414, Oct. 2015, number: 1 Publisher: Nature Publishing Group.
 - [95] E. O’Sullivan-Greene and I. Mareels, “Limits of Observability in Large-Scale Linear Networked Clocks,” *IFAC Proceedings Volumes*, vol. 47, no. 3, pp. 1164–1169, Jan. 2014.
 - [96] E. O’Sullivan-Greene, I. Mareels, L. Kuhlmann, and A. Burkitt, “Observability limits for networked oscillators,” *Automatica*, vol. 50, no. 4, pp. 1087–1099, Apr. 2014.
 - [97] E. O’Sullivan-Greene, L. Kuhlmann, A. Varsavsky, D. B. Grayden, A. N. Burkitt, and I. M. Mareels, “Seizure prediction and observability of EEG sources,” in *Epilepsy: The Intersection of Neurosciences, Biology, Mathematics, Engineering, and Physics*. CRC press, 2011.
 - [98] A. Das, D. Sexton, C. Lainscsek, S. S. Cash, and T. J. Sejnowski, “Characterizing Brain Connectivity From Human Electroencephalography Recordings With Unobserved Inputs During Epileptic Seizures,” *Neural Comput*, vol. 31, no. 7, pp. 1271–1326, Jul. 2019.
 - [99] D. Fan, Z. Yang, C. Yang, Q. Wang, and G. Luan, “Clinically Localized Seizure Focus Maybe Not Exactly the Position of Abating Seizures: A Computational Evidence,” *Nonlinear Dynamics*, vol. 105, no. 2, 2021.
 - [100] J. Friedman, T. J. Hastie, and R. Tibshirani, “Sparse inverse covariance estimation with the graphical lasso,” *Biostatistics (Oxford, England)*, vol. 9, no. 3, pp. 432–41, Jul. 2008.
 - [101] E. Pavez, “Estimation of graph laplacian and covariance matrices,” Phd Thesis, University of Southern California, 2019.
 - [102] X. Gao, B. Xiao, D. Tao, and X. Li, “A survey of graph edit distance,” *Pattern Anal Applic*, vol. 13, no. 1, pp. 113–129, Feb. 2010.
 - [103] M. Roy, S. Schmid, and G. Tredan, “Modeling and measuring graph similarity: The case for centrality distance,” in *Proceedings of the 10th ACM international workshop on Foundations of mobile computing*, 2014, pp. 47–52.

- [104] G. Schalk, D. J. McFarland, T. Hinterberger, N. Birbaumer, and J. R. Wolpaw, “BCI2000: a general-purpose brain-computer interface (BCI) system,” *IEEE Transactions on biomedical engineering*, vol. 51, no. 6, pp. 1034–1043, 2004, publisher: IEEE.
- [105] S. Halder, M. Bensch, J. Mellinger, M. Bogdan, A. Kübler, N. Birbaumer, and W. Rosenstiel, “Online Artifact Removal for Brain-Computer Interfaces Using Support Vector Machines and Blind Source Separation,” *Comput Intell Neurosci*, vol. 2007, 2007.
- [106] J. N. Knight, “Signal fraction analysis and artifact removal in EEG,” PhD Thesis, Colorado State University, 2003.
- [107] A. Pedroni, A. Bahreini, and N. Langer, “Automagic: Standardized preprocessing of big EEG data,” *NeuroImage*, vol. 200, pp. 460–473, Oct. 2019.
- [108] Y. Tanaka, Y. C. Eldar, A. Ortega, and G. Cheung, “Sampling Signals on Graphs: From Theory to Applications,” *arXiv:2003.03957 [cs, eess]*, Jun. 2020, arXiv: 2003.03957.
- [109] A. Sakiyama, Y. Tanaka, T. Tanaka, and A. Ortega, “Efficient sensor position selection using graph signal sampling theory,” in *2016 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Mar. 2016, pp. 6225–6229, iSSN: 2379-190X.
- [110] Z. J. Koles, “The quantitative extraction and topographic mapping of the abnormal components in the clinical EEG,” *Electroencephalography and clinical Neurophysiology*, vol. 79, no. 6, pp. 440–447, 1991, publisher: Elsevier.
- [111] G. Nolte, O. Bai, L. Wheaton, Z. Mari, S. Vorbach, and M. Hallett, “Identifying true brain interaction from EEG data using the imaginary part of coherency,” *Clinical Neurophysiology*, vol. 115, no. 10, pp. 2292–2307, Oct. 2004.
- [112] X. Li, Y. Wu, M. Wei, Y. Guo, Z. Yu, H. Wang, Z. Li, and H. Fan, “A novel index of functional connectivity: phase lag based on Wilcoxon signed rank test,” *Cogn Neurodyn*, vol. 15, no. 4, pp. 621–636, Aug. 2021.
- [113] Q. Zhang, Y. Hu, X. Dong, and X. Feng, “Clinical significance of electroencephalography power spectrum density and functional connection analysis in neonates with hypoxic-ischemic encephalopathy,” *International Journal of Developmental Neuroscience*, vol. 81, no. 2, pp. 142–150, 2021, _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/jdn.10083>.
- [114] J. A. Gaxiola-Tirado, R. Salazar-Varas, and D. Gutiérrez, “Using the Partial Directed Coherence to Assess Functional Connectivity in Electroencephalography Data for Brain-Computer Interfaces,” *IEEE Transactions on Cognitive and Developmental Systems*, vol. 10, no. 3, pp. 776–783, Sep. 2018, conference Name: IEEE Transactions on Cognitive and Developmental Systems.
- [115] B. Blankertz, K.-R. Müller, G. Curio, T. M. Vaughan, G. Schalk, J. R. Wolpaw, A. Schlögl, C. Neuper, G. Pfurtscheller, and T. Hinterberger, “The BCI competition 2003: progress and perspectives in detection and discrimination of EEG single trials,” *IEEE transactions on biomedical engineering*, vol. 51, no. 6, pp. 1044–1051, 2004, publisher: IEEE.
- [116] H. van Dijk, G. van Wingen, D. Denys, S. Olbrich, R. van Ruth, and M. Arns, “The two decades brainclinics research archive for insights in neurophysiology (TDBRAIN) database,” *Sci Data*, vol. 9, no. 1, p. 333, Jun. 2022, number: 1 Publisher: Nature Publishing Group.
- [117] M. Schirner, L. Domide, D. Perdakis, P. Triebkorn, L. Stefanovski, R. Pai, P. Prodan, B. Vaele, J. Palmer, C. Langford, A. Blickensdörfer, M. van der Vlag, S. Diaz-Pier, A. Peyser, W. Klijn, D. Pleiter, A. Nahm, O. Schmid, M. Woodman, L. Zehl, J. Fousek, S. Petkoski, L. Kusch, M. Hashemi, D. Marinazzo, J.-F. Mangin, A. Flöel, S. Akintoye, B. C. Stahl, M. Cepic, E. Johnson, G. Deco, A. R. McIntosh, C. C. Hilgetag, M. Morgan, B. Schuller,

- A. Upton, C. McMurtrie, T. Dickscheid, J. G. Bjaalie, K. Amunts, J. Mersmann, V. Jirsa, and P. Ritter, "Brain simulation as a cloud service: The Virtual Brain on EBRAINS," *NeuroImage*, vol. 251, p. 118973, May 2022.
- [118] P. Sanz Leon, S. A. Knock, M. M. Woodman, L. Domide, J. Mersmann, A. R. McIntosh, and V. Jirsa, "The Virtual Brain: a simulator of primate brain network dynamics," *Front. Neuroinform.*, vol. 7, 2013, publisher: Frontiers.
- [119] M. Coutino, E. Isufi, T. Maehara, and G. Leus, "State-Space Based Network Topology Identification," in *2020 28th European Signal Processing Conference (EUSIPCO)*, Jan. 2021, pp. 1055–1059, iSSN: 2076-1465.
- [120] J. M. Meier, D. Perdikis, A. Blickensdörfer, L. Stefanovski, Q. Liu, O. Maith, H. . Dinkelbach, J. Baladron, F. H. Hamker, and P. Ritter, "Virtual deep brain stimulation: Multiscale co-simulation of a spiking basal ganglia model and a whole-brain mean-field model with The Virtual Brain," *bioRxiv*, p. 2021.05.05.442704, May 2021, publisher: Cold Spring Harbor Laboratory Section: New Results.
- [121] G. Rousselet, H. Isotalus, and S. Henderson, "A dataset for the test-retest reliability assessment of EEG & ERP quantities," Jul. 2018.
- [122] R. G. Andrzejak, K. Lehnertz, F. Mormann, C. Rieke, P. David, and C. E. Elger, "Indications of nonlinear deterministic and finite-dimensional structures in time series of brain electrical activity: Dependence on recording region and brain state," *Physical Review E*, vol. 64, no. 6, p. 061907, 2001, publisher: APS.
- [123] M. Defferrard, L. Martin, R. Pena, and N. Perraudin, "PyGSP: Graph Signal Processing in Python," Oct. 2017.
- [124] M. Otero, P. Prado-Gutiérrez, A. Weinstein, M.-J. Escobar, and W. El-Deredy, "Persistence of EEG Alpha Entrainment Depends on Stimulus Phase at Offset," *Frontiers in Human Neuroscience*, vol. 14, p. 139, 2020, publisher: Frontiers.
- [125] P. Prado-Gutierrez, E. Martínez-Montes, A. Weinstein, and M. Zañartu, "Estimation of auditory steady-state responses based on the averaging of independent EEG epochs," *PLOS ONE*, vol. 14, no. 1, p. e0206018, Jan. 2019.
- [126] P. Prado-Gutiérrez, M. Otero, E. Martínez-Montes, A. Weinstein, M.-J. Escobar, W. El-Deredy, and M. Zañartu, "A method for tracking the time evolution of steady-state evoked potentials," *JoVE (Journal of Visualized Experiments)*, no. 147, p. e59898, 2019.
- [127] A. Veloz, A. Weinstein, S. Pszczolkowski, L. Hernández-García, R. Olivares, R. Muñoz, and C. Taramasco, "Ant Colony Clustering for ROI Identification in Functional Magnetic Resonance Imaging," *Computational Intelligence and Neuroscience*, vol. 2019, 2019, publisher: Hindawi.
- [128] A. Veloz, C. Moraga, A. Weinstein, L. Hernández-García, S. Chabert, R. Salas, R. Riveros, C. Bennett, and H. Allende, "Fuzzy general linear modeling for functional magnetic resonance imaging analysis," *IEEE Transactions on Fuzzy Systems*, vol. 28, no. 1, pp. 100–111, 2019, publisher: IEEE.